

ORIGINAL CONTRIBUTION

Stroke Risk After Bioprosthetic Aortic Valve Replacement in Aortic Stenosis: Systematic Review and Meta-Analysis

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BACKGROUND: Stroke is a possible complication after bioprosthetic aortic valve replacement (AVR) for severe aortic stenosis (AS), impacting morbidity and mortality. Accurate estimates of the proportion of individuals who experience stroke within and beyond the periprocedural period after transcatheter AVR (TAVR), surgical AVR, and valve-in-valve (ViV) replacement are essential for management and prognostication. The objective was to determine the proportion of adults aged >18 who experienced an ischemic stroke after bioprosthetic AVR for AS.

METHODS: A systematic search of MEDLINE, Embase, and Web of Science was conducted from database inception through March 2024. Studies reporting on stroke rates at least 90 days after bioprosthetic AVR for severe AS, including ViV procedures, and meeting predefined eligibility criteria were included. The pooled proportion of individuals experiencing a stroke was estimated for TAVR and ViV procedures, whereas comparative analyses between TAVR and surgical AVR were performed using mixed-effects models in studies directly comparing both procedures.

RESULTS: Twenty-seven studies were included in the native AS treatment cohort, and 5 in the ViV subanalysis. In native AS, the pooled 30-day proportion of individuals who had a stroke after TAVR was 3.0% (95% CI, 2.5–3.9), with different studies reporting major and minor stroke proportions of 1.7% each. At 1 year, all stroke proportion was 5.0% (95% CI, 4.0–6.0), major stroke was 3.0%, and minor stroke was 2.0%. Comparative analysis demonstrated that TAVR was associated with significantly lower odds of all stroke at 30 days compared with surgical AVR (odds ratio, 0.73 [95% CI, 0.57–0.93]). No significant difference in the proportions of individuals who had a stroke was observed in TAVR versus surgical AVR at 1, 2, or 5 years. In the ViV cohort, the pooled 30-day and 1-year all stroke proportion after ViV was 2.0% (95% CI, 1.0–3.0) and 3.0% (95% CI, 2.0–6.0), respectively.

CONCLUSIONS: This meta-analysis provides updated estimates of stroke after bioprosthetic AVR for AS, capturing risk beyond the early periprocedural period. Future studies should investigate the causes of long-term stroke post-AVR, the effects of different antithrombotic therapies on the risk of stroke, as well as the potential impact of these procedures on short and long-term cognitive function.

GRAPHIC ABSTRACT: A graphic abstract is available for this article.

Key Words: aortic valve stenosis ■ heart valves ■ risk ■ transcatheter aortic valve replacement ■ stroke

The rising global life expectancy has led to an increasing prevalence of aortic stenosis (AS), one of the most common valvular heart diseases in the United States

and Europe, and the third most common cardiovascular disease after hypertension and coronary artery disease.^{1–3} Given the lack of effective medical therapies, aortic valve

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Nonstandard Abbreviations and Acronyms

AS	aortic stenosis
AVR	aortic valve replacement
OR	odds ratio
SAVR	surgical aortic valve replacement
TAVR	transcatheter aortic valve replacement
TIA	transient ischemic attack
VARC	Valve Academic Research Consortium
ViV	valve-in-valve

replacement (AVR) remains the only definitive treatment for severe AS.⁴ In the United States, among 246 hospitals performing transcatheter AVR (TAVR), annual AVR volume increased from 19578 in 2008 to 33004 in 2013, reflecting rapid procedural growth during a short period.⁵ However, AVR is associated with possible complications, with stroke representing one of the most serious and clinically significant adverse outcomes. Recent years have seen a notable rise in the use of TAVR among younger patients (<65 years), reflecting its expanding indications. Because younger and lower-risk individuals have longer life expectancy, they face a greater cumulative burden of poststroke morbidity.^{6,7} In addition, the broadening of eligibility criteria and advancements in perioperative care may have increased provider willingness to perform TAVR in older or higher-risk patients, which may further contribute to a rise in stroke rates.⁸ Finally, along with the rise of AVR, the use of valve-in-valve (ViV) TAVR has been increasing in patients with subsequently deteriorated aortic prosthetic valves, as an alternative to redo surgery, and this number is expected to continue to grow.⁹ After AVR, prior reports had suggested a stroke frequency as high as 5.8% at 30 days, and a 30-day mortality of 28.2% in patients with post-AVR-stroke, compared with 6.4% in post-AVR patients without stroke ($P<0.0001$).^{10–12}

Although several observational studies and clinical trials have investigated periprocedural rates of stroke, particularly to compare stroke or thromboembolic events frequency postnative TAVR versus postnative surgical AVR (SAVR), there is a relative scarcity of studies investigating the long-term frequency of stroke after AVR, especially ViV-TAVR.

Within this context, we performed a systematic review and meta-analysis of observational and randomized studies investigating the proportion of individuals with both periprocedural (up to 30 days) and long-term stroke (beyond 30 days) after bioprosthetic AVR for AS of surgical, transcatheter, and ViV transcatheter approaches.

METHODS

This systematic review and meta-analysis were conducted in accordance with the Preferred Reporting Items for Systematic

Reviews and Meta-Analyses guidelines and were prospectively registered in the PROSPERO database (URL: <https://www.crd.york.ac.uk/PROSPERO/>; CRD42024520182). A comprehensive literature search was performed across MEDLINE (via PubMed), EMBASE, and Web of Science, covering all records from database inception through March 2024. The search strategy was designed and executed by a professional medical librarian (S.J.K.), with full details provided in Table S1. Research data are stored in an institutional repository and will be shared on request to the corresponding author.

Study screening and selection were performed independently by 2 reviewers (T.B.D. and S.H.) using the Covidence platform.¹³ Screening occurred in 2 phases: initial title and abstract review, followed by full-text evaluation. Predefined eligibility criteria were applied to identify studies assessing the long-term risk of stroke after bioprosthetic AVR for AS while minimizing confounding from unrelated comorbidities. Eligible studies included randomized controlled trials and prospective observational cohorts enrolling ≥ 50 adults with stenotic, aortic valve disease who underwent surgical or transcatheter bioprosthetic AVR and reported on stroke frequency beyond 90 days postprocedure. To reduce stroke risk attributable to other cardioembolic causes and to reduce heterogeneity of stroke risk in the study population, studies were excluded if they included mechanical valve implantation, native aortic insufficiency or regurgitation, redo surgeries, patients with left ventricular ejection fraction <30%, advanced kidney disease, bicuspid aortic valve, active infective endocarditis, or aortic root replacement.¹⁴ Studies limited to early postoperative stroke (<90 days) were also excluded. ViV procedures were included and analyzed separately as a prespecified subgroup.

Stroke outcomes were categorized as major (fatal or disabling) and minor (non disabling), based on the definitions provided in each study. When multiple publications stemmed from the same cohort, the most comprehensive report, defined by sample size and follow-up duration, was selected. Data were extracted from the earliest baseline to the latest follow-up available. Complete eligibility criteria are detailed in Table S2. Risk of bias was assessed using the Cochrane Risk of Bias 2 tool for randomized controlled trials and the Newcastle-Ottawa Scale for observational and registry studies,^{15,16} following standardized domain-based evaluation criteria to ensure methodological consistency across study designs (Tables S3 and S4).

For each study, the proportion of individuals who had a stroke was calculated as the number of individuals with stroke events divided by the total number of individuals undergoing AVR. They were reported at multiple time points, with longer-term estimates reflecting cumulative events up to each respective follow-up period. Pooled proportions of individuals experiencing stroke and 95% CI were estimated using binomial-normal models, that is, a logistic regression model with random intercept. Prediction intervals were also reported to additionally quantify the distribution of stroke proportion for future, similar studies. For studies directly comparing TAVR and SAVR, mixed-effects conditional logistic regression models were used to estimate odds ratios (ORs) of stroke outcomes. Heterogeneity was assessed using I^2 and τ^2 , which are statistical measures used to quantify the degree of variability among study estimates beyond chance. All analyses were performed in R 4.4.1 (R Core Team, 2024).

RESULTS

The initial database search yielded 3001 studies, of which 1932 were eligible for screening. First phase screening yielded 483 articles selected for full-text screening. After the application of predefined inclusion and exclusion criteria (Table S2), 451 studies were excluded. A total of 32

studies met eligibility, and after detailed classification, 27 studies^{10,17–42} were included in the native AS cohort and 5 in the ViV sub-analysis^{43–47} (Figure 1). Data for several cohorts were derived from multiple published studies on the same population.^{48–56}

Among observational and registry studies assessed with the Newcastle-Ottawa Scale, most showed low to

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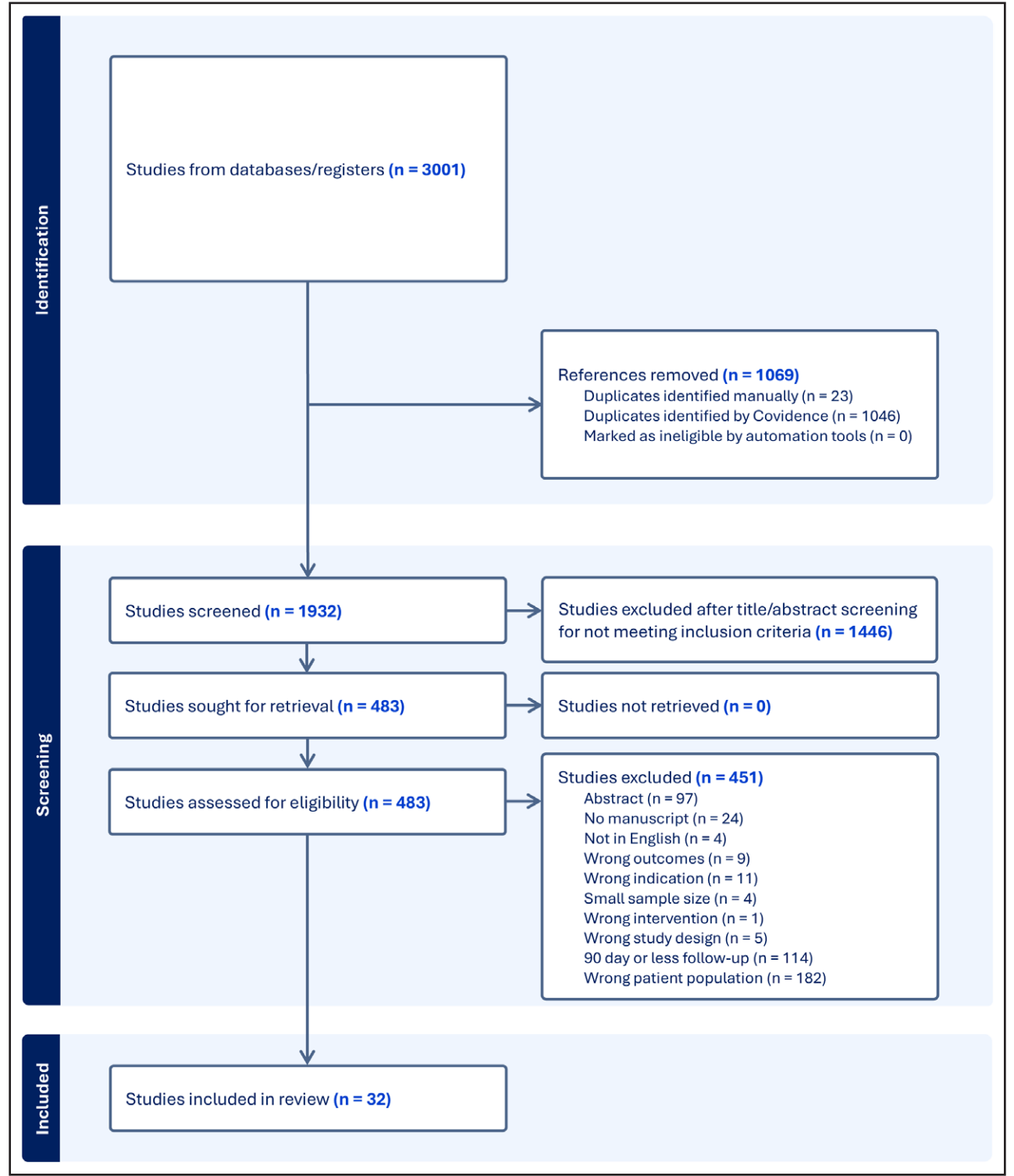


Figure 1. Study selection.

moderate risk of bias, with limitations primarily in cohort comparability. All randomized controlled trials, evaluated using the Cochrane Risk of Bias 2 tool, were considered to have low overall risk of bias, with only isolated domains showing some concerns. Collectively, these findings indicate that the overall methodological quality of included studies was acceptable, supporting confidence in the pooled estimates (Tables S3 and S4).

In the treatment of the native severe AS group, the 27 included studies comprised 15 252 patients. The mean age across cohorts was 80.7 ± 2.97 years, with females representing 48.8% of the population. Baseline characteristics are presented in Table S5. At 30 days, post-procedure stroke was reported in 17 studies evaluating TAVR. The pooled 30-day proportion of individuals who had a stroke was 3.0% (95% CI, 2.5–3.9), with moderate between-study heterogeneity ($P=60.5\%$, $\tau^2=0.11$). Major stroke at 30 days was reported in 15 studies, yielding a pooled proportion of 1.7% ([95% CI, 1.0–2.0]; $P=48.1\%$, $\tau^2=0.11$), whereas the pooled estimate for minor stroke from 12 studies was 1.7% ([95% CI, 1.3–2.1]; $P=16.0\%$, $\tau^2=0.03$). These values do not sum to the overall stroke estimate because not all studies reported outcomes stratified by stroke severity. Four studies reported 30-day transient ischemic attack (TIA) rates, with a pooled estimate of 0.5% ([95% CI, 0.3–1.0]; Table 1).

At 1-year follow-up, 14 studies reported stroke outcomes after TAVR. The pooled 1-year proportion of individuals who had a stroke was 5.0% (95% CI, 4.0–6.0), with substantial heterogeneity across studies ($P=75.0\%$, $\tau^2=0.15$; Figure 2A). Major stroke at 1 year, reported in 12 studies, had a pooled proportion of 3.0% ([95% CI, 2.0–4.0]; $P=73.6\%$, $\tau^2=0.22$; Figure 2B), while the minor stroke proportion was estimated at 2.0% ([95% CI, 1.5–2.7]; $P=55.3\%$, $\tau^2=0.13$; Figure 2C). The proportion of individuals with TIA at 1 year, reported in 2 studies, was 1.4% ([95% CI, 1.0–1.9]).

Five studies compared stroke outcomes between TAVR and SAVR. At 30 days, TAVR was associated with significantly lower odds of all stroke (OR, 0.73 [95% CI, 0.57–0.93]; Figure 3A) and major stroke

(OR, 0.65 [95% CI, 0.44–0.95]). At later time points, no significant differences were observed between TAVR and SAVR. For all stroke, the ORs were 0.83 (95% CI, 0.67–1.02) at 1 year (Figure 3B), 0.85 (95% CI, 0.68–1.06) at 2 years (Figure 3C), and 1.01 (95% CI, 0.85–1.19) at 5 years (Figure 3D). For major stroke, ORs were 0.79 (95% CI, 0.60–1.05) at 1 year, 0.84 (95% CI, 0.65–1.09) at 2 years, and 0.99 (95% CI, 0.80–1.23) at 5 years. For minor stroke, the respective ORs were 0.95 (95% CI, 0.58–1.54), 0.89 (95% CI, 0.66–1.19), and 1.03 (95% CI, 0.81–1.31). However, the odds of TIA at 2 years were significantly higher in patients undergoing TAVR compared with SAVR (OR, 1.49 [95% CI, 1.06–2.11]). At 5 years, TIA remained numerically higher in the TAVR group (OR, 1.34 [95% CI, 0.99–1.81]), although the difference was not statistically significant (Table 2).

In the ViV subanalysis, 5 studies met the inclusion criteria.^{43–47} In all included studies, patients initially underwent SAVR and later received transcatheter ViV for treatment of their failed surgical bioprosthesis. The mean age of patients undergoing ViV procedures was 77.5 years, and the average age of the degenerated bioprosthesis was 10.18 years. Indications for ViV were bioprosthetic stenosis in 55.4% of cases, regurgitation in 24.2%, and mixed lesions in 19.5%. Baseline characteristics are summarized in Table S6. The pooled 30-day proportion of all stroke in the ViV cohort was 2.0% (95% CI, 1.0–3.0), with major stroke estimated at 0.4% (95% CI, 0.1–1.5) and minor stroke at 1.0% (95% CI, 0.2–5.0%). At 1 year, the pooled proportion of all stroke was 3.0% (95% CI, 2.0–6.0; Table 3; Figure 4). Key comparative stroke outcomes across valve replacement modalities and time points are summarized in Table 4.

DISCUSSION

The proportion of individuals who have a stroke after bioprosthetic AVR in AS remains significant, warranting continued clinical attention and additional research to minimize the risk. In the perioperative and postoperative

Table 1. Proportions of Individuals With Stroke and TIA at 30 Days and 1-Year Time Points After Transcatheter Aortic Valve Replacement

Outcomes	Time point	No. of studies	Pooled proportion (95% CI)	95% prediction interval	<i>P</i>	τ^2
All stroke	30 d	17	0.03 (0.025–0.039)	0.02–0.06	60.5%	0.11
Major stroke	30 d	15	0.017 (0.01–0.02)	0.01–0.03	48.1%	0.11
Minor stroke	30 d	12	0.017 (0.013–0.021)	0.01–0.03	16.0%	0.03
TIA	30 d	4	0.005 (0.003–0.01)	0.003,0.01	0%	0
All stroke	1 y	14	0.05 (0.04–0.06)	0.02–0.10	75.0%	0.15
Major stroke	1 y	12	0.03 (0.02–0.04)	0.01–0.07	73.6%	0.22
Minor stroke	1 y	13	0.02 (0.015–0.027)	0.01–0.04	55.3%	0.13
TIA	1 y	2	0.014 (0.01–0.019)	0.01–0.02	0%	0

TIA indicates transient ischemic attack.

settings, stroke is a known major contributor to adverse outcomes such as prolonged hospitalization, morbidity, and in-hospital mortality.⁵⁷ In a study of patients undergoing valve surgery, those who developed a stroke had an in-hospital mortality rate of 24% compared with 4.6% in those without stroke.¹⁴ Similarly, in

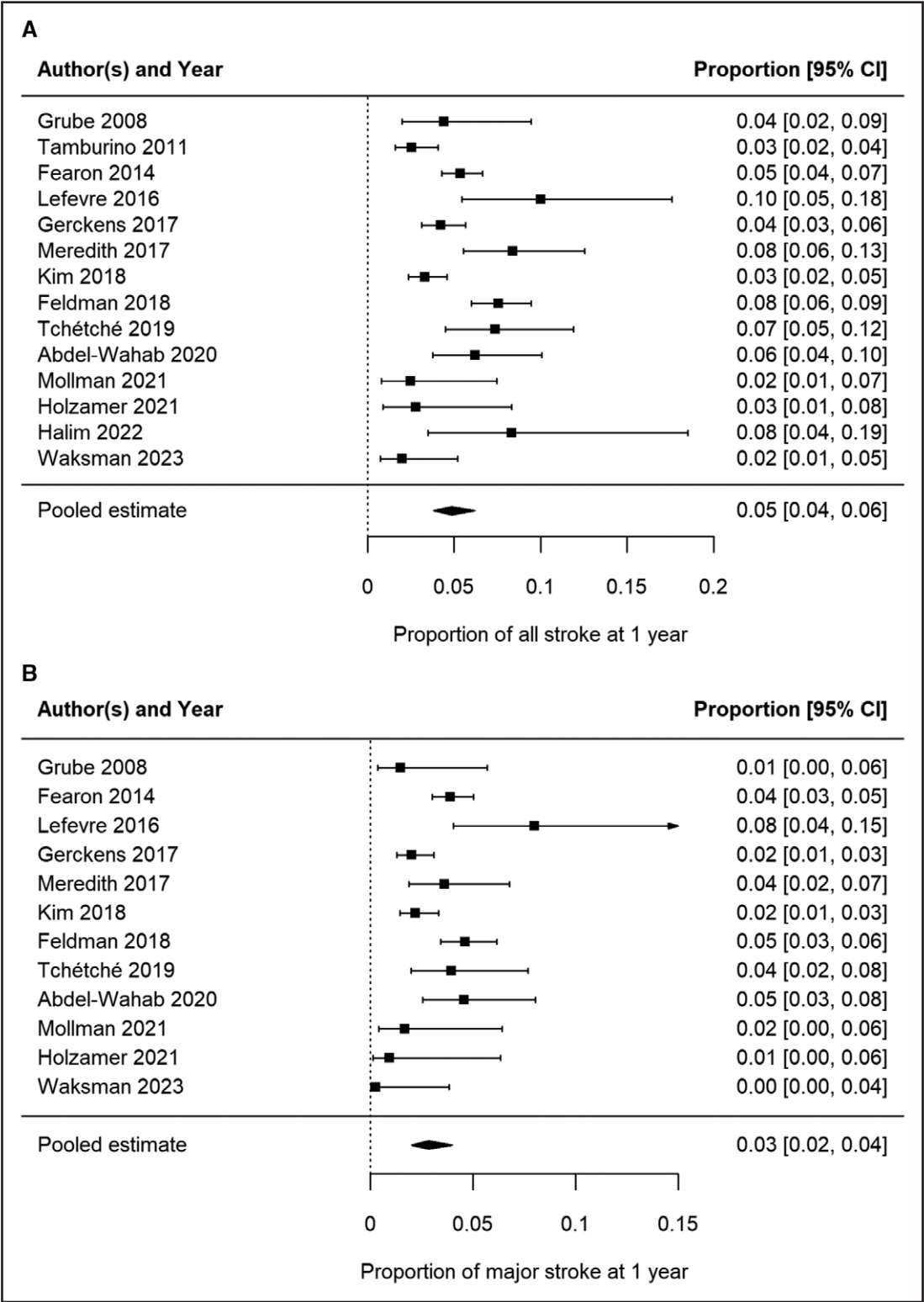


Figure 2. xxx.
A, Proportion of individuals with any stroke after transient ischemic attack (TAVR) at 1 year. **B**, Proportion of individuals with major stroke after TAVR at 1 year. **C**, Proportion of individuals with minor stroke after TAVR at 1 year. (Continued)

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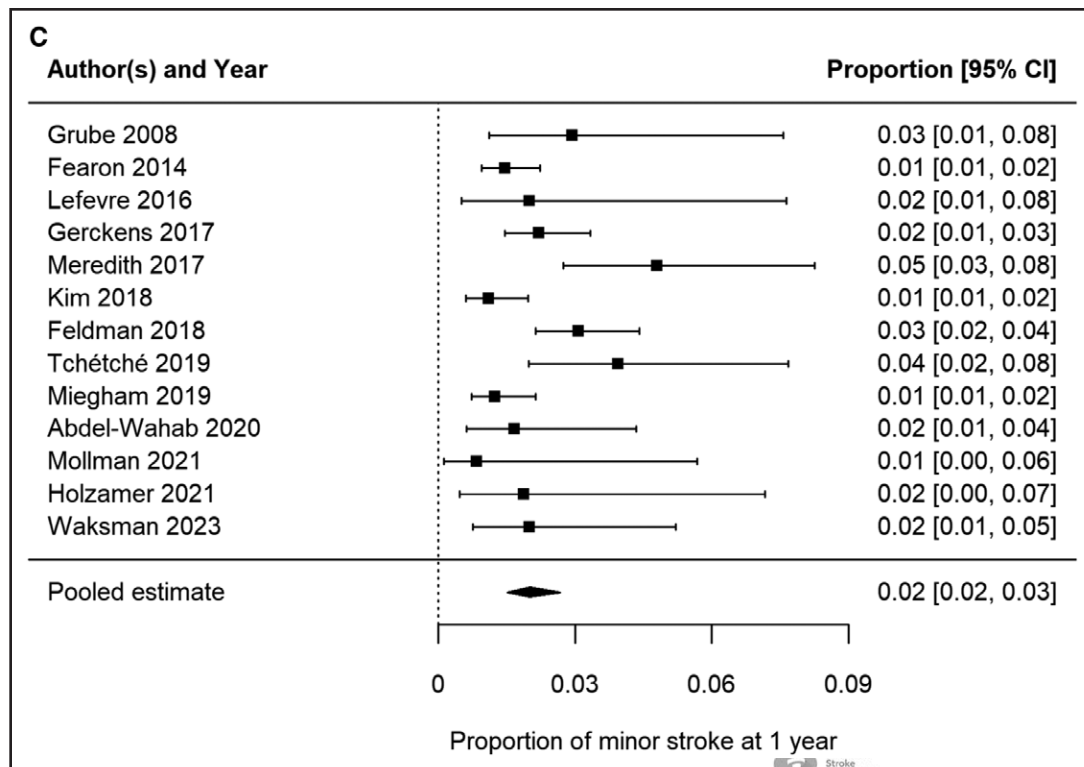


Figure 2 Continued.

a prospective cohort study, Messe et al⁵⁸ reported that stroke was associated with a more than 9-fold increase in in-hospital mortality among AVR patients. To date, the causes underlying perioperative stroke in patients after bioprosthetic AVR remain understudied but may be related to cardiac manipulation and cardioembolism. Although some studies suggest that manipulation of the aortic root or the use of large delivery systems may dislodge atherosclerotic debris and contribute to cerebral embolization in the short-term, particularly within the first 7 days, current evidence is mixed. Overall, studies have not shown a consistent reduction in periprocedural stroke with cerebral embolic protection, with outcomes possibly influenced by operator and institutional learning.^{59–62} Wang and Phan⁶³ reported that although the use of cerebral embolic protection devices during TAVR was associated with a lower rate of stroke compared with TAVR alone, this difference did not reach statistical significance. Similarly, in the PROTECTED TAVR trial, the incidence of any stroke within 72 hours or before discharge was not significantly different between those who received cerebral embolic protection devices and control groups (2.3% versus 2.9%; $P=0.30$). However, when evaluating only disabling stroke, a significant reduction was observed with cerebral protection use (0.5% versus 1.3% [95% CI, -1.5 to -0.1]).⁶⁴ Accordingly, the proposed mechanisms of periprocedural stroke after AVR remain inconclusive.^{59–62} In addition, a meta-analysis comparing transfemoral and transapical

approaches for AVR found no significant difference in 30-day stroke rates between the 2 access routes.⁶⁵

For strokes after bioprosthetic AVR beyond the periprocedural period, the causes of stroke are not well understood, with a paucity of studies characterizing the cause. This meta-analysis highlights the need for deeper investigation into the etiological landscape of stroke post-AVR. Also, the studies included in our meta-analysis did not stratify patients based on their antithrombotic regimens, as these were often not mentioned in the individual studies, limiting our ability to assess how specific therapies influence stroke risk. In addition, none of the included trials discussed recurrent stroke events, precluding further interpretation.

Proportion of Individuals With Stroke at 30 Days

We found that the 30-day stroke proportion after TAVR was 3%, which is consistent with those of Huded et al⁶⁶ who reported a 2.3% stroke rate at 30 days post-TAVR, a prior meta-analysis that estimated a 2.9% early stroke rate,⁶⁷ and an analysis comparing transapical and transfemoral access approaches that found 30-day stroke rates of 3.4% and 3.8%, respectively.⁶⁵ Together, these findings underscore a consistent 30-day stroke proportion of $\approx 2.3\%$ to 3.8% after TAVR. About half of these strokes were classified as major, consistent with findings by Eggebrecht et al,⁶⁸ who reported a 2.5% frequency of major stroke within the first 30 days post-TAVR.

Proportion of Individuals With Stroke at 1 Year

At 1 year after TAVR, the pooled proportion of stroke was 5.0% (95% CI, 4.0%–6.0%) and heterogeneity was substantial ($P=75.0\%$; $\tau^2=0.15$). By comparison, in the

community-based Framingham cohort, adults aged 80 to 84 years—which includes the mean age of our study population—had 10-year stroke probabilities of 22% to 24%, corresponding to annualized incidence rates of ≈ 25 to 27 per 1000 person-years ($\approx 2\%$ – 3% per year). Thus,

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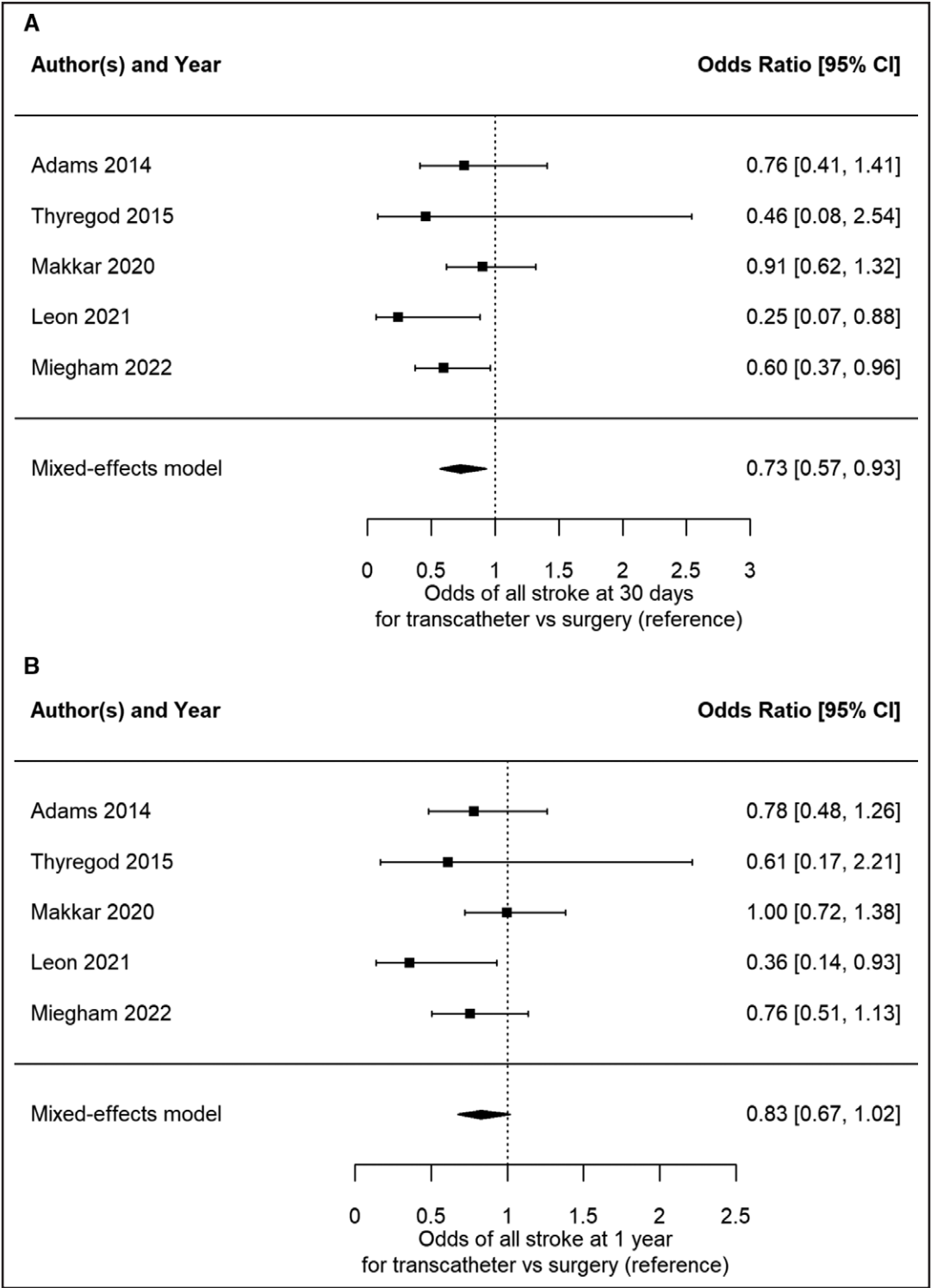


Figure 3. xxx. **A**, Odds of stroke in transient ischemic attack (TAVR) vs surgical aortic valve replacement (SAVR) at 30 days. **B**, Odds of stroke in TAVR vs SAVR at 1 year. **C**, Odds of stroke in TAVR vs SAVR at 2 years. **D**, Odds of stroke in TAVR vs SAVR at 5 years. (Continued)

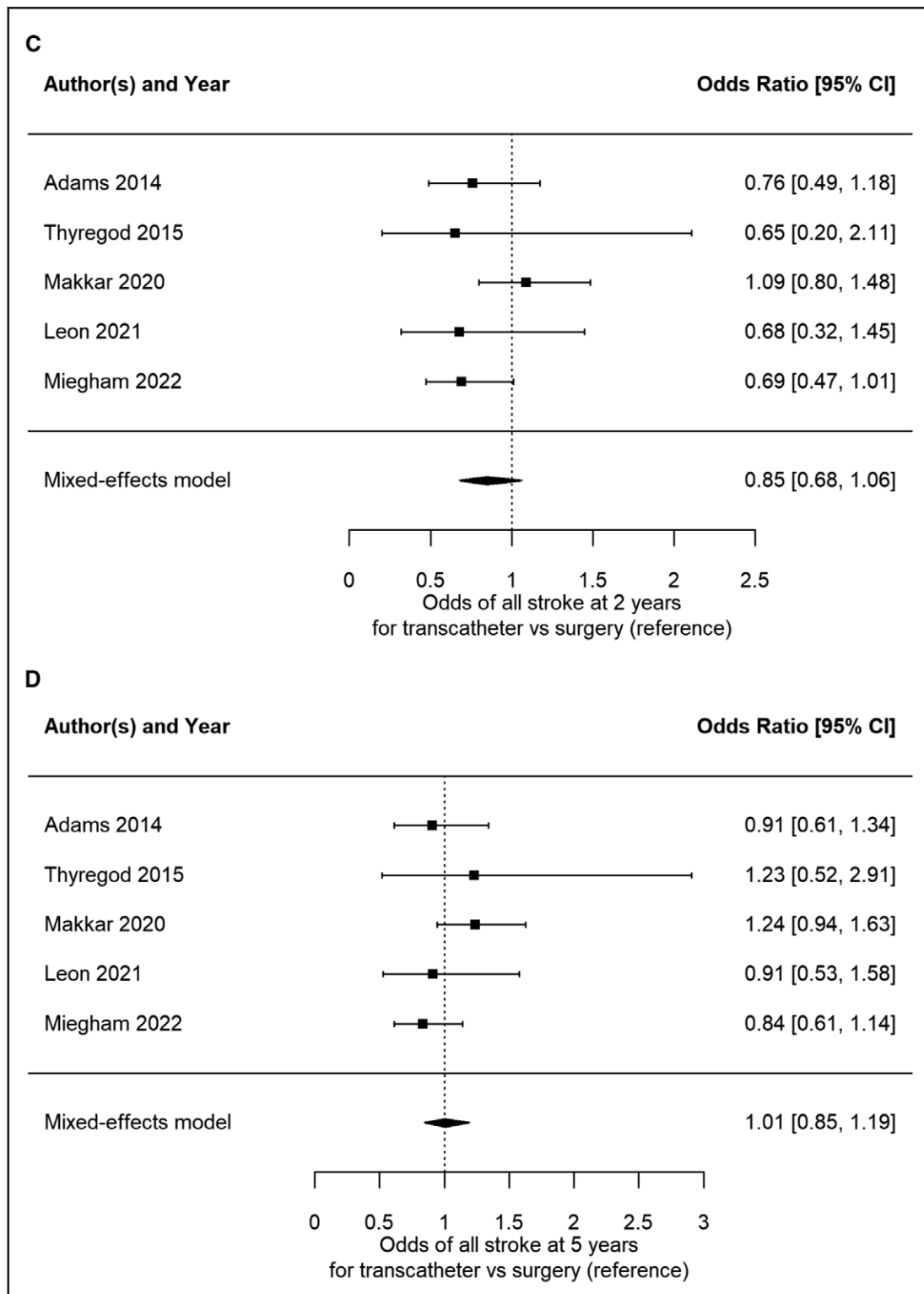


Figure 3 Continued.

the risk of stroke at 1 year after TAVR appears nearly double that expected in an age-matched general population cohort.^{69,70} Heterogeneity among studies could possibly be explained by differences in patient risk factors and comorbidities, patient selection, concomitant procedures

and procedural techniques, and study designs (observational versus randomized clinical trials). Importantly, the CI remained relatively narrow, suggesting a stable and clinically meaningful estimate across varied populations. These findings underscore that a proportion of 5%

Table 2. Stroke and TIAs in TAVR vs SAVR

Outcomes	Time point	No. of studies	OR (95% CI) Transcatheter vs surgery (ref)	P	τ^2
All stroke	30 d	5	0.73 (0.57–0.93)	0.0%	0.000
Major stroke	30 d	3	0.65 (0.44–0.95)	0.0%	0.000
Minor stroke	30 d	3	0.80 (0.47–1.36)	25.6%	0.050
TIA	30 d	3	1.76 (0.90–3.45)	0.0%	0.000
All stroke	1 y	4	0.83 (0.67–1.02)	0.0%	0.000
Major stroke	1 y	4	0.79 (0.60–1.05)	0.0%	0.000
Minor stroke	1 y	4	0.95 (0.58–1.54)	43.9%	0.100
TIA	1 y	3	1.50 (0.97–2.30)	0.0%	0.000
All stroke	2 y	5	0.85 (0.68–1.06)	11.7%	0.008
Major stroke	2 y	4	0.84 (0.65–1.09)	0.0%	0.000
Minor stroke	2 y	4	0.89 (0.66–1.19)	0.0%	0.000
TIA	2 y	4	1.49 (1.06–2.11)	0.0%	0.000
All stroke	5 y	5	1.01 (0.85–1.19)	6.7%	0.003
Major stroke	5 y	4	0.99 (0.80–1.23)	0.0%	0.000
Minor stroke	5 y	4	1.03 (0.81–1.31)	0.0%	0.000
TIA	5 y	3	1.34 (0.99–1.81)	0.0%	0.000

OR indicates odds ratio; SAVR, surgical aortic valve replacement; TAVR, transcatheter aortic valve replacement; and TIA, transient ischemic attack.

represents a moderate risk with substantial clinical implications. Stroke severity analysis further revealed a greater burden of disabling events, with major strokes occurring at a pooled estimate of 3% (95% CI, 2.0%–4.0%) compared with 2.0% (95% CI, 1.5%–2.7%) for minor strokes. This imbalance highlights the disproportionate contribution of major strokes to long-term morbidity and functional impairment in this population. These findings are in line with previous literature, which reported a combined 12-month frequency of stroke and TIA of 5.2%, with the majority of events (6 out of 7) classified as major strokes.⁶⁸ Taken together, our results reinforce the importance of cerebrovascular risk as a persistent and clinically relevant complication after TAVR, obviating the need to further understand the cause of strokes in this setting and developing targeted prevention strategies.

TAVR Versus SAVR

TAVR has recently been preferred to SAVR in the setting of bioprosthetic heart valves in patients deemed

inoperable or at high surgical risk. In our analysis, TAVR was associated with a significantly lower risk of stroke at 30 days compared with SAVR, particularly for major stroke. However, this early benefit did not persist over time, as no significant differences in stroke rates between TAVR and SAVR, whether all stroke, major, or minor, were observed at 1, 2, or 5 years. At 30 days, prior studies comparing TAVR and SAVR outcomes have reported variable stroke rates. In the CoreValve trial and a meta-analysis, rates were numerically lower with TAVR (4.9% versus 6.2%, $P=0.46$; and 2.7% versus 3.1%, $P=0.08$), whereas the PARTNER trial showed higher rates with TAVR (3.8% versus 2.1%; $P=0.20$). Major stroke rates at 30 days in another meta-analysis were also comparable and nonsignificant (2.5% versus 2.9%; $P=0.89$).

Among the limited number of studies that have investigated long-term stroke outcomes, at 1 year, total stroke rates remained nonsignificant across CoreValve (8.8% versus 12.6%; $P=0.10$), the meta-analysis (5.0% versus 4.6%, $P=0.96$), and PARTNER (5.1% versus 2.4%; $P=0.07$), with similar findings for major stroke (eg, 2.1% versus 4.6%; $P=0.71$).^{18,71,72} In addition, Talanas et al⁷³ reported no significant difference in the frequency of long-term disabling stroke between TAVR and SAVR ($P=0.07$). Other studies likewise found no significant difference in disabling stroke at the longest available follow-up (6.8% versus 7.2%; $P=0.71$).⁷⁴ Therefore, based on our study and consistent with other studies, the stroke rates after the first 30 days of the procedure are not significantly different between TAVR and SAVR.

Valve-in-Valve

Over the past decade, the use of bioprosthetic aortic valves has increased significantly, primarily due to their advantage of obviating the need for long-term anticoagulation compared with mechanical valves.^{75–77} However, bioprosthetic valves carry their own limitations. Structural valve deterioration is a known complication, typically presenting as either stenosis—most commonly due to progressive calcification—or, less frequently, pannus formation or thrombosis. Regurgitation, on the other hand, may result from leaflet wear and tear, calcification, or endocarditis. Long-term durability remains a concern, with failure rates reported in 20% to 30% of valves by 10 years and up to 50% by 15 years postimplantation.⁷⁸

Table 3. Proportions of Individuals With Stroke After Valve-in-Valve Bioprostheses

Outcomes	Time point	Number of studies	Pooled proportion (95% CI)	95% prediction interval	P	τ^2
All stroke	30 d	4	0.02 (0.01–0.03)	0.01–0.03	0.0%	0
Major stroke	30 d	3	0.004 (0.001–0.015)	0.001–0.015	0.0%	0
Minor stroke	30 d	3	0.01 (0.002–0.05)	0.001–0.09	52.0%	0.86
All stroke	1 y	5	0.03 (0.02–0.06)	0.02–0.07	33.2%	0.09
Major stroke	1 y	3	0.01 (0.005–0.025)	0.005–0.025	0.0%	0

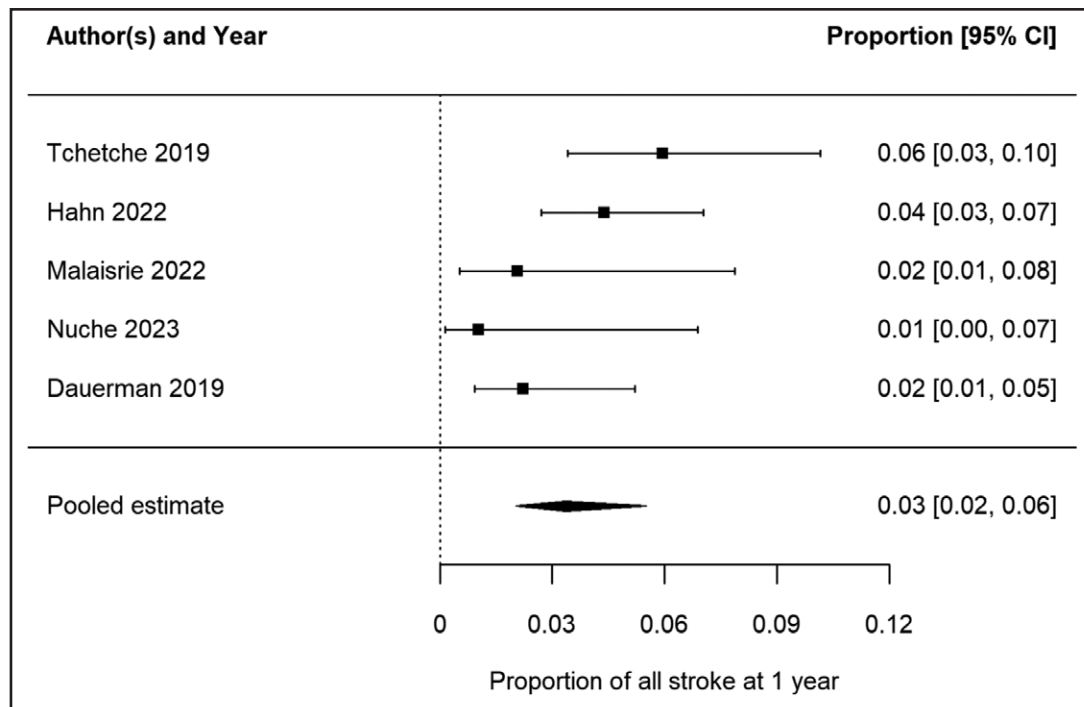


Figure 4. Proportion of individuals with stroke after valve-in-valve at 1 year.



These durability concerns, coupled with the growing prevalence of bioprosthetic valves, have expanded the indications for TAVR beyond native valve disease to include failed bioprostheses, giving rise to the ViV-TAVR technique.⁷⁹ In our meta-analysis, the mean time to bioprosthetic degeneration was 10.18 years, aligning with the previously reported durability window of 10 to 15 years.⁷⁸ Indications for ViV intervention were bioprosthetic stenosis in 55.4% of cases, regurgitation in 24.2%, and mixed pathology in 19.5%. These distributions are broadly consistent with data from the ViV Registry,⁸⁰ which reported stenosis in 42%, regurgitation in 34%, and mixed pathology in 24%. The pooled proportion of individuals experiencing a stroke in our analysis was 2.0% at 30 days and 3% at 1 year. Our 30-day estimate of stroke post-ViV-TAVR mirrors the 2.2% estimate reported by a previous meta-analysis by Macherey et al.⁸¹ Notably, we did not identify previous studies that have provided pooled estimates of stroke at 1 year or beyond for ViV-TAVR. A natural benchmark for ViV-TAVR is redo SAVR. Kaneko et al⁸² analyzing the Society of Thoracic Surgeons Adult Cardiac Surgery Database, reported a postoperative stroke rate of 1.9% after redo SAVR. In addition, 3 separate meta-analyses consistently found no significant short-term difference in stroke risk between ViV-TAVR and redo SAVR: 1.5% versus 2.6%,⁸³ 0.8% versus 1%,⁸⁴ and 2.0% in both groups.⁸⁵ Extending to long-term outcomes, Nagasaka et al⁸⁶ likewise reported similar rates, with both procedures having a stroke incidence of 5.2% at 2 years. Our findings therefore suggest that, in the short term, stroke rates after ViV-TAVR are

consistent with native TAVR and comparable with redo SAVR. However, the long-term risk of stroke after ViV-TAVR remains underexplored and warrants further investigation. Collectively, these results reinforce the safety profile of ViV-TAVR with respect to stroke occurrence and support its role as a viable alternative to redo surgery, particularly in patients at elevated surgical risk.

Strengths and Contributions of This Meta-Analysis

This meta-analysis provides an updated and comprehensive evaluation of stroke risk in patients undergoing bioprosthetic valve replacement, a population that continues to expand due to broader indications and evolving procedural technologies. Our analysis focuses on long-term stroke rates after both TAVR and SAVR, utilizing contemporary data that reflect current clinical practice. Notably, this is the first meta-analysis, to our knowledge, to report pooled long-term stroke risk estimates in patients undergoing ViV procedures, a subgroup of increasing relevance with the growing adoption of TAVR for structural valve deterioration.

Although prior observational studies, randomized trials, and meta-analyses have examined stroke in this context, existing literature is limited by heterogeneity in study design, inconsistent follow-up intervals, and variable definitions and reporting of stroke outcomes. By applying standardized methodology, stringent selection criteria, and conducting time-specific analyses, our study addresses these limitations and offers clinically

Table 4. Comparative Summary of the Proportion of Individuals With Stroke After Bioprosthetic Aortic Valve Replacement

Time point	Stroke outcome	TAVR (native AS) pooled proportion % (95% CI)	TAVR vs SAVR OR (95% CI)	ViV-TAVR pooled proportion % (95% CI)	Clinical interpretation
30 d	All stroke	3.0 (2.5–3.9)	0.73 (0.57–0.93)	2.0 (1.0–3.0)	Less strokes in TAVR compared with SAVR in the periprocedural period
	Major stroke	1.7 (1.0–2.0)	0.65 (0.44–0.95)	0.4 (0.1–1.5)	Less stroke in TAVR compared with SAVR in the periprocedural period
	Minor stroke	1.7 (1.3–2.1)	0.80 (0.47–1.36)	1.0 (0.2–5.0)	No significant difference in the proportion of individuals with stroke between TAVR and SAVR in the periprocedural period
	TIA	0.5 (0.3–1.0)	1.76 (0.9–3.45)	...	No significant difference in the proportion of individuals experiencing a TIA after TAVR compared with SAVR in the periprocedural period
1 y	All stroke	5.0 (4.0–6.0)	0.83 (0.67–1.02)	3.0 (2.0–6.0)	The proportion of individuals experiencing a stroke after TAVR increases over time; no significant difference between TAVR and SAVR at 1 y postprocedure
	Major stroke	3.0 (2.0–4.0)	0.79 (0.60–1.05)	1.0 (0.5–2.5)	The proportion of individuals experiencing a disabling stroke is similar between TAVR and SAVR
	Minor stroke	2.0 (1.5–2.7)	0.95 (0.58–1.54)	–	The proportion of individuals experiencing a minor stroke is similar between TAVR and SAVR
	TIA	1.4 (1.0–1.9)	1.50 (0.97–2.30)	...	The proportion of individuals experiencing a TIA is similar between TAVR and SAVR
2 y	All stroke	...	0.85 (0.68–1.06)	...	The proportion of individuals experiencing a stroke is similar between TAVR and SAVR at 2 y
	Major stroke	...	0.84 (0.65–1.09)	...	The proportion of individuals experiencing a disabling stroke is similar between TAVR and SAVR at 2 y
	TIA	...	1.49 (1.06–2.11)	...	The proportion of individuals experiencing a TIA is higher in TAVR compared with SAVR at 2 y
5 y	All stroke	...	1.01 (0.85–1.19)	...	The proportion of individuals experiencing a stroke is similar between TAVR and SAVR at 5 y
	Major stroke	...	0.99 (0.80–1.23)	...	The proportion of individuals experiencing a disabling stroke is similar between TAVR and SAVR at 5 y
	TIA	...	1.34 (0.99–1.81)	...	Trend toward higher TIA with TAVR, not statistically significant

AS indicates aortic stenosis; OR, odds ratio; SAVR, surgical aortic valve replacement; TAVR, transcatheter aortic valve replacement; TIA, transient ischemic attack; and ViV, valve-in-valve.

meaningful pooled estimates—particularly at the 1-year mark and beyond. In addition, we highlight a key gap in the literature: the lack of consistent reporting on anti-thrombotic therapy or stroke cause or characterization across studies. This represents an important area for future research, as specific antithrombotic strategies may play a critical role in reducing stroke risk in this high-risk population.

Limitations

Despite rigorous methodology and inclusion of high-quality studies, heterogeneity in outcome definitions, including variability in the criteria used to define stroke (eg, VARC [Valve Academic Research Consortium]-1,⁸⁷ VARC-2,⁸⁸ VARC-3,⁸⁹ or investigator-defined definitions) and in the grading severity (major, minor), limits the precision of pooled estimates (Table S7). In addition, the analysis for stroke post-ViV-TAVR was limited by the small number of studies evaluating the ViV procedure. Most trials did not apply uniform neurological follow-up or short-term systematic neuroimaging, and few incorporated formal neurocognitive assessments, leaving clinically silent or subtle cognitive postprocedural deficits

potentially undetected and underreported. Moreover, substantial variation exists in baseline patient characteristics across trials, but this is an inherent limitation to conducting meta-analyses. Information on prior stroke events or history of atrial fibrillation were infrequently reported, and when reported were heterogeneous. None of the included studies stratified stroke outcomes by sex, precluding assessment of potential sex-based differences in risk. In addition, operator experience and procedural techniques (eg, use of cerebral embolic protection or access route) evolved over time but were not consistently stratified in the source data for stroke within 30 days, and therefore, we could not conduct analyses to look for differences in stroke rates based on use of cerebral embolic protection devices. Long-term data remain sparse, particularly for next-generation TAVR platforms and ViV procedures. Insufficient reporting of antithrombotic regimens limited comparisons by treatment strategy, and the absence of patient-level data prevented adjustment for key confounding variables. An additional limitation is that at the 2- and 5-year time points for long-term stroke risk, the included studies reported outcomes only as intention-to-treat percentages without providing the number of patients remaining under observation.

Consequently, the exact sample size at later time points could not be consistently extracted, and readers should interpret long-term pooled outcomes with awareness of this reporting constraint. The absence of consistent cerebrovascular follow-up in the patients undergoing AVR also makes TIAs, minor, and silent strokes underrepresented. Furthermore, we are unable to comment on the potential mechanisms underlying the strokes in this population, as this information was not provided in the published studies included in our analysis. Finally, we could not comment or analyze differences in individual-level risk factors for those with stroke versus not post-AVR, as this information was also often not included in published studies.

Conclusions

This meta-analysis provides an updated short and long-term assessment of stroke risk in a growing population of individuals undergoing bioprosthetic AVR for AS. It also underscores that despite advances in surgical and transcatheter techniques, the long-term risk of stroke remains clinically significant and a serious long-term concern in patients receiving bioprosthetic valves. Further research is needed to develop a greater understanding of the incidence of stroke post-ViV beyond the periprocedural period, mechanisms underlying stroke in this population, the influence of different antithrombotic therapies on the risk of stroke, as well as the potential impact of these procedures on short and long-term cognitive function.

ARTICLE INFORMATION

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Author Contributions

Dr Bou Dargham designed the protocol, performed article screening, data extraction, synthesis, and interpretation, and drafted the manuscript. Dr Hassani contributed to article screening, data extraction, the refinement of inclusion and exclusion criteria, and drafting the manuscript. T. Truong performed the statistical analyses and contributed to the interpretation of results. Dr Kaplan developed and executed the search strategy. B.C. Mac Grory, Dr Daubert, Dr Ryan, Dr Moumneh, and Dr Feng contributed to study design discussions and manuscript drafting. All authors critically reviewed and approved the final manuscript. Dr El Hussein conceived the study, designed the protocol, drafted the manuscript, provided senior supervision, methodological guidance, and assisted in the development of the analytical plan.

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Supplemental Material

Tables S1–S7
PRISMA Checklist

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